

# Optimization and Numerical Methods

## Weekly assignments

### 1 Assignment 1 (Due on Saturday 8 Feb 2025. Topics: Convex set)

1. Let  $x$  be a real-valued random variable with  $\text{prob}(x = a_i) = p_i, i = 1, \dots, n$ , where  $a_1 < a_2 < \dots < a_n$ . Now,  $p \in \mathbb{R}^n$  lies in the standard probability simplex  $P = \{p | 1^T p = 1, p \geq 0\}$ . Which of the following conditions are convex in  $p$ ? (That is, for which of the following conditions is the set of  $p \in P$  that satisfy the condition convex?).
  - (a)  $\alpha \leq Ef(x) \leq \beta$ , where  $Ef(x)$  is the expected value of  $f(x)$ , i.e.,  $Ef(x) = \sum_{i=1}^n p_i f(a_i)$ . (The function  $f: \mathbb{R} \rightarrow \mathbb{R}$  is given.)
  - (b)  $\text{prob}(x > \alpha) \leq \beta$ .
  - (c)  $Ex^2 \leq \alpha$ .
  - (d)  $Ex^2 \geq \alpha$ .
  - (e)  $\text{var}(x) \leq \alpha$ , where  $\text{var}(x) = E(x - Ex)^2$  is the variance of  $x$ .
  - (f)  $\text{var}(x) \geq \alpha$ .
  - (g)  $\text{quartile}(x) \geq \alpha$ , where  $\text{quartile}(x) = \inf\{\beta \mid \text{prob}(x \leq \beta) \geq 0.25\}$ .
  - (h)  $\text{quartile}(x) \leq \alpha$ .
2.
  - (a) Prove that the intersection of two convex sets is also a convex set.
  - (b) Show that if  $S_1$  and  $S_2$  are two convex sets in  $\mathbb{R}^{m+n}$ , then so is their partial sum  $S = \{(x, y_1 + y_2) : x \in \mathbb{R}^m, y_1, y_2 \in \mathbb{R}^n, (x, y_1) \in S_1, (x, y_2) \in S_2\}$ .